



Indian Institute of Technology Kanpur  
Department of Mathematics and Statistics  
**Introduction to Bayesian Analysis (MTH422)**  
Quiz 3, Date: March 24, 2025, Monday

Time: 4:00–4:40 PM (4:00–4:55 PM for DAP)

Spring 2025

Max point: 10

A

**There will NOT be any partial credit in any question.** All assumptions (whether observations are IID or not, independent or not) should be explicitly and correctly written.

The following is a JAGS snippet and outputs. Answer the questions on the next page based on these codes and outputs.

```
1 data <- list(Y = Y, n = length(Y))
2 model_string <- textConnection("model{
3 for(i in 1:n){
4 Y[i] ~ dnorm(X[i], 100)
5 }
6 for(i in 1:n){
7 X[i] ~ dnorm(mu, tau)
8 }
9 tau ~ dgamma(0.01, 0.01)
10 sigma <- 1/sqrt(tau)
11 mu ~ dnorm(0, 0.0001)
12 }")
13
14 model <- jags.model(model_string, data = data, n.chains = 1, quiet = TRUE)
15 update(model, 10000, progress.bar = "none")
16
17 params <- c("mu", "sigma")
18 samples <- coda.samples(model, variable.names = params,
19                          n.iter = 20000, thin = 2, progress.bar = "none")
20
21 summary(samples)
22
23 1. Empirical mean and standard deviation for each variable,
24    plus standard error of the mean:
25
26      Mean      SD Naive SE Time-series SE
27 mu      2.0433 0.02748 0.0002748      0.0002644
28 sigma 0.2572 0.02127 0.0002127      0.0002285
29
30 2. Quantiles for each variable:
31
32      2.5%   25%   50%   75%  97.5%
33 mu      1.9890 2.025 2.0433 2.0616 2.0980
34 sigma 0.2192 0.242 0.2563 0.2712 0.3015
35
36 geweke.diag(samples) # based on default setting
37      mu      sigma
38 0.5217 1.1945
```

1. Write down the two layers of the random effect model and the two priors. All parameters and hyperparameters (parameters of the prior), independence or IID-type conditions should be explicit. ( $0.5 \times 4 = 2$  points)
2. What are the burn-in length, post-burn-in length before thinning, post-burn-in length after thinning? How many samples are used for drawing inference here? ( $0.5 \times 4 = 2$  points)
3. What is the dimension of the posterior distribution in this problem? Write down the purpose of Line 17 of the snippet in this context. ( $0.5 \times 2 = 1$  point)
4. Calculate the effective sample size for  $\sigma$ . (1 point)
5. Report the Monte Carlo estimates of the 95% equal-tailed posterior credible interval of  $\mu$  and  $\sigma$ . ( $0.5 \times 2 = 1$  point)
6. The Monte Carlo estimate of the posterior probability of  $\mu$  being within the interval  $(2, 2.05)$  is at least 0.25. Justify. (1 point)
7. Instead of the posterior of  $\sigma$ , if our interest lies in the posterior of  $\sigma^2$ , how would you modify the above code? Further, if you want to obtain the inference about  $\mu + 3\sigma$ , how would you modify the above code? ( $0.5 \times 2 = 1$  point)
8. What can you conclude from the Geweke statistics reported here? Calculate the Gelman-Rubin diagnostic here. If you cannot calculate, justify why. ( $0.5 \times 2 = 1$  point)